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Generalized Beam Theory with Exact Periodic Boundary Conditions for the Buckling Analysis of Omega-Stringer-Stiffened Composite Panels

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Abstract: Stringer-stiffened composite panels in aircraft covers carry many identical stringers, so that their buckling behaviour is commonly analysed on one representative cell whose side edges satisfy periodic boundary conditions. Closed-form models with periodic conditions rely on idealisations of the skin–stringer interaction, while shell finite element models with periodic constraint equations are accurate but expensive and give no insight into the deformation mechanisms. Generalized Beam Theory (GBT) provides such insight at low cost, but its cross-section analysis has been developed for members with free longitudinal edges or closed contours, and periodic side edges have not been treated. This study combines GBT and the Ritz method into a shear-deformable GBT–Ritz formulation for omega-stringer-stiffened composite panels in which the periodic condition is built into every family of deformation modes: the conventional modes through constraint rows that are projected out family by family, the warping functions through a constrained Vlasov least-squares problem, and the shear and transverse-shear modes through merging of the nodal functions of the two side edges. Every mode, and therefore every truncated mode basis, satisfies the periodic condition exactly. The walls are laminated Reissner–Mindlin plates with full membrane–bending and anisotropic coupling, the stringer is tied to the skin with its true offset, the pre-buckling state is the first-order solution under the actual end loads, and the amplitude functions are cubic B-splines, which cover simply supported and clamped loaded edges. The modal matrices are given explicitly in the notation of classical GBT, and the formulation reduces to the classical GBT buckling equations for uncoupled walls under uniform compression. For 30 panels with simply supported loaded edges (five lay-ups, two stringer pitches, three aspect ratios), the critical loads agree with Abaqus shell models with periodic constraint equations within 0.57 % on average and 1.25 % at most; for the 30 panels with clamped loaded edges, they agree with an independent spline finite strip model within 0.30 % on average and 1.17 % at most. An energy-based mode decomposition identifies all critical modes as local skin buckling with global and distortional contributions below 4 %, at a computing time of a few seconds per case.

Keywords: generalized beam theory; Ritz method; periodic boundary conditions; stiffened panels; omega stringer; laminated composites; buckling

1. Introduction

Stringer-stiffened panels are the standard construction of aircraft fuselage and wing covers, and their buckling behaviour is often the governing design criterion of weight-critical laminated composite structures [1,2]. Among the stiffener shapes, the omega (hat) stringer is attractive because its closed cell provides high bending and torsional stiffness, a wide bonding area and good damage tolerance [3,4]. A typical cover carries many identical stringers. Its buckling behaviour away from the free or supported longitudinal edges is therefore represented by one repeating cell, provided that the side edges of this cell are given the conditions that the neighbouring cells impose, namely periodic boundary conditions [5–7].

The treatment of the repeating cell is a classical topic of the analysis of stiffened panels. In exact and finite strip methods for prismatic plate assemblies, the infinitely wide panel was analysed early through its repeating portion: the exact strip programs VIPASA and VICONOPT treat infinitely wide or long panels by repeating a portion and its supports [8,9], and the finite strip method in general handles such assemblies with a small number of strips [10,11]. The underlying principle is the theory of waves in periodic structures (Bloch–Floquet theory), by which the response of an infinite periodic structure follows from one cell with phase-related boundary conditions [12].

At the Institute for Lightweight Engineering and Structural Mechanics (LSM), a series of closed-form and semi-analytical models for stringer-stiffened composite panels with periodic boundary conditions has been developed. Mittelstedt derived closed-form buckling loads of blade-stringer-stiffened composite plates considering periodic boundary conditions [5] and extended the approach to the stiffener–plate interaction in

local buckling [6]. For omega stringers, Mittelstedt and Beerhorst presented a closed-form buckling analysis [3], and Mittelstedt and Schröder treated the local postbuckling of hat-stringer-stiffened plates [13]. Schilling and Mittelstedt developed a closed-form approximate solution for the local buckling of omega-stringer-stiffened panels [4], a column-buckling model with periodic boundary conditions [7], and models for shear buckling, compression–shear interaction and local postbuckling [14–16]. Most recently, El Yaakoubi-Mesbah and Mittelstedt presented a closed-form analytical solution for the local buckling of omega-stringer-stiffened composite panels under compression [17], a computational model for local buckling [18] and a closed-form postbuckling analysis including the stringer-foot effects [19]; the stability of stiffened laminates is summarised in [20]. These models are efficient and transparent, but they rest on idealisations: the stringer is replaced by elastic restraints, the skin is reduced to one or a few plate strips with assumed shape functions, local and global buckling are usually treated by separate models, and the pre-buckling stress is taken as uniform. The membrane–bending coupling of unsymmetric laminates is typically included through a reduced bending stiffness [21]. The physics-informed neural network of the authors for the local buckling of omega-stringer-stiffened panels [22] was built on such a reduced energy model. Shell finite element models with periodic constraint equations between the nodes of the two side edges, on the other hand, are accurate and general, but each design requires a new mesh and a large eigenvalue analysis, and the buckling mode is obtained only as a nodal field, without a decomposition into mechanically meaningful deformation mechanisms.

Generalized Beam Theory (GBT), originated by Schardt [23,24] and developed further by Davies and Leach [25], describes the deformation of a prismatic thin-walled member as a superposition of cross-section deformation modes whose amplitudes vary along the member axis. A cross-section analysis determines the modes and the associated modal matrices, and a member analysis determines the amplitude functions. The modal decomposition separates global, distortional, local and shear-related deformations and has made GBT one of the most transparent tools for coupled instabilities of thin-walled members, as summarised in the state-of-the-art reports by Camotim et al. [26] and Mittelstedt [27,28]. Silvestre and Camotim formulated first- and second-order GBT for arbitrary orthotropic materials [29,30], and later shear-deformable GBT for composite members and its vibration counterpart [31,32]. Dinis et al. extended the cross-section analysis to arbitrarily branched open sections [33]. Gonçalves et al. introduced a new approach to the calculation of the cross-section deformation modes [34], and Bebiano et al. rationalised the cross-section analysis into natural, shear and transverse-extension mode families [35], together with the shear deformation modes [36], implemented in the code GBTUL 2.0 [37]. Closed sections were treated, among others, for circular cylindrical shells and tubes [38] and for regular polygonal tubes [39]. For laminated composites, GBT has been applied to FRP open-section columns [40] and to laminated CFRP cylindrical stiffened panels [41], and Taig and Ranzi formulated GBT for stiffened sections [42].

The Ritz method provides the complementary tool for the member analysis: the continuous buckling problem is replaced by a finite-dimensional one over admissible functions that satisfy the essential boundary conditions [43], and its efficiency depends on a basis that reflects the expected solution [44]. Outside GBT, Kühn et al. used such variational discrete-plate models for the local buckling of shear-deformable laminated beams [45]. Kharghani and Mittelstedt combined GBT and the Ritz method into a GBT–Ritz approach: for global, local and distortional buckling of shear-deformable composite beams with open cross-sections [46], for composite box girders [47], and for thin-walled composite beams with unsymmetric layups, for which the membrane–bending coupling is consistently included [48,49]. In this approach GBT defines the kinematics through the modes, and the Ritz method determines how the modal amplitudes vary along the member.

Two gaps motivate the present study. The first gap lies in GBT. GBT has been developed for members: open sections with free longitudinal edges, branched sections, and closed contours. The skin of a stiffened panel cell, however, ends at two side edges that are neither free nor closed: the displacement field at one side edge must equal that at the other. This condition has not been formulated in GBT, and it cannot simply be imposed afterwards on a set of classical modes. Classical mode generation starts from a few natural nodes, whereas the periodic condition couples the two ends of the open contour; a truncated set of classical modes then contains too few modes that satisfy it, and the rigid-body and warping modes of the open cell are not admissible for the periodic cell. In addition, the omega stringer forms closed cells with the skin, so that Vlasov's condition of null membrane shear strain cannot be fulfilled by the warping functions, the stringer is offset from the skin by the half thicknesses, and the pre-buckling state is not uniform near the loaded edges, where the load enters the stringer through the eccentric foot. The second gap lies in the treatment of periodic conditions. The closed-form models are efficient, but they rely on idealisations of the skin–stringer interaction, separate local and global buckling, and do not account for the non-uniform pre-buckling state; shell finite element models do not have these restrictions, but they are expensive for parametric studies and give no modal insight; exact and finite strip methods treat the repeating portion efficiently, but they are

formulated in nodal degrees of freedom and do not provide the decomposition into global, distortional and local mechanisms.

GBT and periodic boundary conditions complement each other. The periodic cell is prismatic, so that the member analysis of GBT applies directly; the modal decomposition identifies the buckling mechanisms of the cell; and the reduced number of generalised coordinates makes parametric studies cheap. The present study combines them. Its novelty is the formulation of the periodic condition within the cross-section analysis of GBT, such that every deformation mode, and therefore every combination and truncation of the modes, satisfies it exactly, together with a shear-deformable GBT–Ritz member analysis of omega-stringer-stiffened panels with symmetric and unsymmetric laminates. The modal matrices are given explicitly in the notation of classical GBT and of the GBT–Ritz approach [49], and the formulation is shown to reduce to the classical GBT buckling equations for uncoupled walls under uniform compression. The formulation is verified against Abaqus shell models with periodic constraint equations for simply supported loaded edges and against an independent spline finite strip model for clamped loaded edges.

The paper is organised as follows. Section 2 develops the formulation: the cross-section analysis of the periodic cell, the periodic boundary conditions, the member analysis by the Ritz method and the mode decomposition. Section 3 presents and discusses the results, and Section 4 summarises the conclusions.

2. Theory and formulations

GBT describes the deformation of a prismatic thin-walled member by cross-section deformation modes whose amplitudes vary along the member axis. The analysis is divided into two stages [24,29,35]: a cross-section analysis, which determines the deformation modes and the modal matrices, and a member analysis, which determines the amplitude functions. Based on the strain–displacement relations, the strain energy of the cross-section is established in terms of the modal amplitudes, and the governing equations follow from the stationarity of the total potential energy, which is evaluated here by the Ritz method, as in the GBT–Ritz approach [46,49]. In the present work the cross-section analysis is extended by the periodic boundary conditions of the panel cell (Section 2.2).

2.1. Cross-section analysis

The panel is the periodic cell of an infinitely wide panel with identical omega stringers at the pitch b_s (Figure 1a). The cell contains two stringers and has the width $W = 2b_s$ and the length $L = \beta b_s$, with the aspect ratio β. The global cross-section axes are X across the panel and Y normal to the skin towards the stringers; x is the longitudinal axis along the stringers. The skin occupies $-W \le X \le 0$, and its side edges $X = -W$ and $X = 0$ lie in the middle of a skin bay. The mid-line of the cross-section is divided into walls (skin, feet, webs, crowns) and the walls into segments of length at most $b_s/25$, with at least two segments per wall. As in classical GBT, the end points of the walls are the natural nodes and the remaining segment end points are intermediate nodes (Figure 1b) [24,35]. For $b_s = 150$ mm the cell has 92 nodes, 19 walls and 93 segments. The coordinate s runs along the mid-line of every wall, z is the coordinate through the thickness, and u, v and w are the displacements in the directions x, s and z. The two side nodes of the skin are denoted s_L ($X = -W$) and s_R ($X = 0$).

The analysis begins with the displacement field of shear-deformable GBT [31,46],

$$u(x,s) = \sum_i \bar{u}_i(s)\,\varphi_{i,x}(x) + \sum_j \bar{u}_j(s)\,\varphi_j(x), \tag{1}$$

$$v(x,s) = \sum_i \bar{v}_i(s)\,\varphi_i(x), \tag{2}$$

$$w(x,s) = \sum_i \bar{w}_i(s)\,\varphi_i(x), \tag{3}$$

which is extended here by the transverse shear rotations of the Reissner–Mindlin walls,

$$\beta_x(x,s) = \sum_r \bar{c}_r(s)\,\chi_r(x), \qquad \beta_s(x,s) = \sum_r \bar{b}_r(s)\,\vartheta_r(x). \tag{4}$$

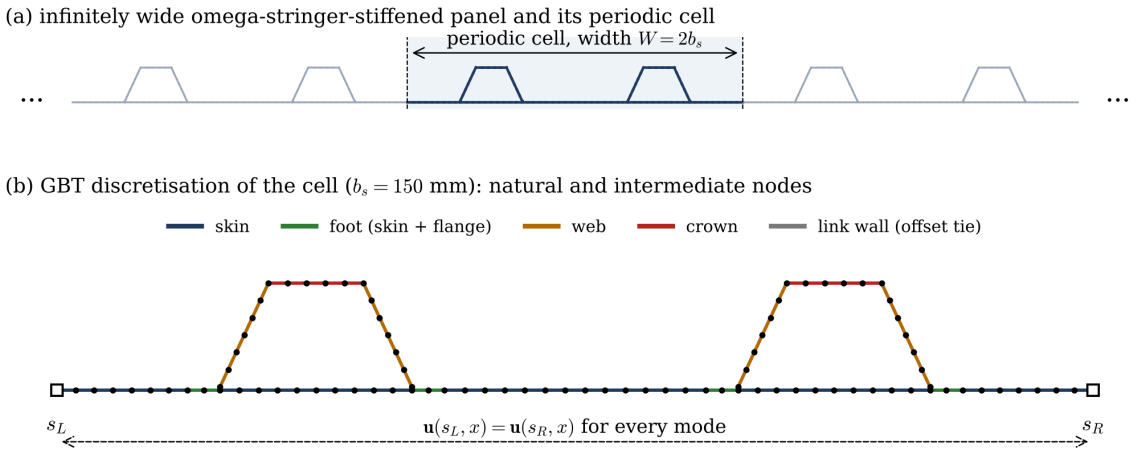


Figure 1. (a) Infinitely wide omega-stringer-stiffened panel and its periodic cell of width $W = 2b_s$. (b) GBT discretisation of the cell for $b_s = 150$ mm with natural and intermediate nodes; skin, feet (skin and flange as one laminate), webs, crowns and link walls of the offset tie; s_L and s_R are the side nodes on which the periodic condition acts.

The index i refers to the conventional deformation modes, j to the shear modes and r to the transverse shear (Mindlin) modes. The functions $\bar{u}_i(s)$, $\bar{v}_i(s)$ and $\bar{w}_i(s)$ describe the displacements of a conventional mode in the axial, tangential and out-of-plane directions, $\bar{u}_j(s)$ is the warping of a shear mode, and $\bar{c}_r(s)$ and $\bar{b}_r(s)$ are the distributions of the shear rotations. The corresponding amplitude functions are $\varphi_i(x)$, $\varphi_j(x)$, $\chi_r(x)$ and $\vartheta_r(x)$. The uniform axial displacement is included among the shear modes, with $\bar{u}_j = 1$.

The rotations of the wall normal are $\phi_x = -w_{,x} + \beta_x$ and $\phi_s = -w_{,s} + \beta_s$, so that the strains at a distance z from the mid-surface are

$$\varepsilon_{xx} = u_{,x} + z\,\kappa_{xx}, \qquad \varepsilon_{ss} = v_{,s} + z\,\kappa_{ss}, \qquad \gamma_{xs} = u_{,s} + v_{,x} + z\,\kappa_{xs}, \tag{5}$$

with the curvatures and the transverse shear strains

$$\begin{gathered} \kappa_{xx} = -w_{,xx} + \beta_{x,x}, \qquad \kappa_{ss} = -w_{,ss} + \beta_{s,s}, \qquad \kappa_{xs} = -2w_{,xs} + \beta_{x,s} + \beta_{s,x}, \\ \gamma_{xz} = w_{,x} + \phi_x = \beta_x, \qquad \gamma_{sz} = w_{,s} + \phi_s = \beta_s. \end{gathered} \tag{6}$$

For $\beta_x = \beta_s = 0$, Eqs. (5) and (6) are the strain–displacement relations of the GBT–Ritz approach for thin walls, $\varepsilon_{xx} = u_{,x} - z\,w_{,xx}$, $\varepsilon_{ss} = v_{,s} - z\,w_{,ss}$ and $\gamma_{xs} = u_{,s} + v_{,x} - 2z\,w_{,xs}$ [49].

In every ply p, Hooke's law relates the in-plane stresses to the strains by the transformed reduced stiffness matrix $\bar{\mathbf{Q}}^{(p)}$ $(i, j = 1, 2, 6)$, and the transverse shear stresses to γ_{xz}, γ_{sz} by $\bar{\mathbf{Q}}_s^{(p)}$. Integration through the thickness gives the laminate stiffnesses [50–52]

$$\left(A_{ij}, B_{ij}, D_{ij}\right) = \sum_{p=1}^{N_L} \int_{z_p}^{z_{p+1}} \bar{Q}_{ij}^{(p)} \left(1, z, z^2\right) dz, \qquad \mathbf{A}_s = \frac{5}{6} \sum_{p=1}^{N_L} \bar{\mathbf{Q}}_s^{(p)} \left(z_{p+1} - z_p\right), \tag{7}$$

with the number of plies N_L, and the constitutive law of a wall reads

$$\hat{\sigma} = \mathbf{C}\,\hat{\varepsilon}, \qquad \mathbf{C} = \begin{bmatrix} \mathbf{A} & \mathbf{B} & \mathbf{0} \\ \mathbf{B} & \mathbf{D} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_s \end{bmatrix}, \tag{8}$$

with the stress resultants $\hat{\sigma} = [N_{xx}, N_{ss}, N_{xs}, M_{xx}, M_{ss}, M_{xs}, Q_x, Q_s]^{\mathsf{T}}$ and the generalised strains $\hat{\varepsilon} = [u_{,x}, v_{,s}, u_{,s} + v_{,x}, \kappa_{xx}, \kappa_{ss}, \kappa_{xs}, \gamma_{xz}, \gamma_{sz}]^{\mathsf{T}}$. All entries of **A**, **B** and **D** are retained, including the membrane–bending coupling **B** of unsymmetric laminates and the anisotropic terms A_{16}, A_{26}, D_{16}, D_{26}. The transverse

membrane resultant N_{ss} is not condensed out, because the transverse-extension modes carry the strain $v_{,s}$ explicitly.

A stringer foot consists of the skin and the stringer flange, which are tied in the shell model. In GBT the two layers form one wall referred to the skin mid-plane. With $\mathbf{A}_0$, $\mathbf{B}_0$, $\mathbf{D}_0$ of the combined stack about its own mid-plane and z_f the distance from the skin mid-plane to the mid-plane of the stack, the parallel-axis transformation gives

$$\mathbf{A}_f = \mathbf{A}_0, \qquad \mathbf{B}_f = \mathbf{B}_0 + z_f\,\mathbf{A}_0, \qquad \mathbf{D}_f = \mathbf{D}_0 + 2z_f\,\mathbf{B}_0 + z_f^2\,\mathbf{A}_0. \tag{9}$$

In the shell model, the stringer mid-surface is tied to the skin with the offset $c = (t_s + t_f)/2$, where t_s and t_f are the thicknesses of skin and stringer. In GBT every web root is a node at $Y = c$, joined to the skin node below it by a link wall of height c. The link wall must transmit the in-plane displacements and the rotation like a line of rigid offset links, but it must not add axial or twisting stiffness. Its laminate is built from the skin stiffnesses $\mathbf{A}_0$, $\mathbf{D}_0$ and $\mathbf{A}_{s0}$ as

$$\begin{gathered} A^L_{11} = 10^{-3}A_{0,11}, \quad A^L_{22} = k_L A_{0,22}, \quad A^L_{66} = k_L A_{0,66}, \\ D^L_{11} = 10^{-3}D_{0,11}, \quad D^L_{22} = k_L D_{0,22}, \quad D^L_{66} = 10^{-3}D_{0,66}, \quad \mathbf{A}^L_s = k_L \mathbf{A}_{s0}, \end{gathered} \tag{10}$$

with $k_L = 10^3$ and all other entries zero; link walls carry no load. The results do not depend on k_L, whereas a link wall with a stiff twisting term D^L_{66} would lock the rotation of the web root and over-predict the critical load (Section 3.5); omitting it is the consistent representation of a row of rigid links.

Within every segment p of length b_p between the nodes q and q', with $\eta = (s - S_q)/b_p \in [0, 1]$, the functions $\bar{u}$ and $\bar{v}$ vary linearly, so that they are described by their nodal values. The transverse displacement $\bar{w}$ is the cubic Hermite interpolation of the nodal normal displacements $\bar{w}_q$, $\bar{w}_{q'}$ and the nodal rotations θ_q, $\theta_{q'}$ ($\theta = \bar{w}_{,s}$):

$$\begin{gathered} \bar{w}(\eta) = H_1(\eta)\,\bar{w}_q + b_p H_2(\eta)\,\theta_q + H_3(\eta)\,\bar{w}_{q'} + b_p H_4(\eta)\,\theta_{q'}, \\ H_1 = 1 - 3\eta^2 + 2\eta^3, \quad H_2 = \eta - 2\eta^2 + \eta^3, \quad H_3 = 3\eta^2 - 2\eta^3, \quad H_4 = -\eta^2 + \eta^3. \end{gathered} \tag{11}$$

The rotation is shared by all walls that meet at a node (rigid fold lines). The in-plane displacement of node q is the vector $\mathbf{U}_q = (U_{X,q}, U_{Y,q})$; on a segment with unit tangent $\mathbf{t}$ and unit normal $\mathbf{n}$, $\bar{v} = \mathbf{t}\cdot\mathbf{U}$ and $\bar{w} = \mathbf{n}\cdot\mathbf{U}$ at its end nodes. The space spanned by Eq. (11) is that of the classical description by nodal transverse moments, which the GBT–Ritz approach uses with the segment bending stiffness D_{22} [46,49]; the nodal form of Eq. (11) is used here because it turns the periodic conditions of Section 2.2 into linear relations between nodal quantities.

Let N be the number of nodes. The in-plane nodal displacements $\mathbf{U} = [\mathbf{U}_1; \ldots; \mathbf{U}_N] \in \mathbb{R}^{2N}$ and the nodal rotations $\Theta \in \mathbb{R}^N$ describe the in-plane deformation of the cross-section. With the matrix $\mathbf{E}_\ell$ that gives the elongation of every segment, $(\mathbf{E}_\ell \mathbf{U})_p = \mathbf{t}_p \cdot (\mathbf{U}_{q'} - \mathbf{U}_q)$, and following the organisation of the mode families in [35,36], the conventional modes are defined by the families

- **G (global):** rigid in-plane translations and rotation of the cross-section, as far as the side conditions admit them;
- **D (distortional):** in-plane nodal displacements without segment elongation, $\mathbf{E}_\ell \mathbf{U} = \mathbf{0}$ (Vlasov's null transverse membrane strain), orthogonal to G and L;
- **L (local):** a unit normal displacement of one node that is not a wall junction, and a pure rotation of one node (the in-plane fields of G, D and T carry no nodal rotation, so that the nodal rotations are independent coordinates);
- **T (transverse extension):** segment elongation patterns, the columns of $\mathbf{E}_\ell^{\mathsf{T}}$;

and the further modes by

- **S (shear):** the nodal hat functions $h_q(s)$ of the warping $\bar{u}_j$, one of which is replaced by the uniform axial mode $\bar{u}_j = 1$;
- **M (Mindlin):** for β_x, a nodal vector Ω_q in the plane of the cross-section, with $\bar{c}_r = \mathbf{t}\cdot\Omega_q$ interpolated linearly on every segment (one vector at a node inside a straight wall, two at a fold line); for β_s, one nodal value $\bar{b}_r$ per node, interpolated linearly.

Since $\mathbb{R}^{2N} = \ker \mathbf{E}_\ell \oplus \text{range } \mathbf{E}_\ell^\mathsf{T}$, the families G, D, L and T together span all in-plane nodal displacements, and the rotation modes make the nodal rotations independent. The conventional modes are therefore a complete nodal description of the in-plane deformation of the cross-section, organised in the GBT families.

The warping $\bar{u}_i$ of the G and D modes follows in classical GBT from Vlasov's condition of null membrane shear strain, $\bar{u}_{i,s} + \bar{v}_i = 0$ on every wall [24,53]. The omega stringer and the skin form closed cells, in which this condition cannot be satisfied exactly. The warping is therefore determined as the stiffness-weighted least-squares solution

$$\min_{\bar{\mathbf{u}}} \sum_p \hat{A}_{66,p}\, b_p \left(\frac{\bar{u}_{q'} - \bar{u}_q}{b_p} + \bar{v}_p\big|_{\eta=1/2} \right)^2 \quad \text{subject to} \quad \mathbf{c}_{ax}^\mathsf{T} \bar{\mathbf{u}} = 0, \tag{12}$$

where the sum runs over all segments, $\hat{A}_{66} = A_{66} - A_{26}^2/A_{22}$ is the membrane shear stiffness for $N_{ss} = 0$, and $c_{ax,q} = \sum_{p \ni q} \frac{1}{2} \hat{A}_{11,p}\, b_p$ with $\hat{A}_{11} = A_{11} - A_{12}^2/A_{22}$, so that the stiffness-weighted mean warping vanishes (no axial content). For an open section, Eq. (12) reproduces the classical warping with zero residual; in the closed cells, the remaining membrane shear strain is carried by the shear modes S.

Inserting Eqs. (1)–(4) into Eqs. (5) and (6), the generalised strains of a mode become products of a cross-section function and an amplitude function or one of its derivatives (Table 1). The conventional entries are those of classical GBT, the entry $v_{,s} = \bar{v}_{i,s}\varphi_i$ belongs to the transverse extension, and the entries of the shear and Mindlin modes are those of shear-deformable GBT.

Table 1. Non-zero generalised strains of one mode, multiplied by the amplitude function (φ), its first derivative (φ^I) or its second derivative (φ^{II}).

Mode (amplitude)	φ	φ^I	φ^{II}
conventional (φ_i)	$v_{,s} = \bar{v}_{i,s}$, $\kappa_{ss} = -\bar{w}_{i,ss}$	$\gamma_{xs} = \bar{u}_{i,s} + \bar{v}_i$, $\kappa_{xs} = -2\bar{w}_{i,s}$	$u_{,x} = \bar{u}_i$, $\kappa_{xx} = -\bar{w}_i$
shear (φ_j)	$\gamma_{xs} = \bar{u}_{j,s}$	$u_{,x} = \bar{u}_j$	–
Mindlin β_x (χ_r)	$\kappa_{xs} = \bar{c}_{r,s}$, $\gamma_{xz} = \bar{c}_r$	$\kappa_{xx} = \bar{c}_r$	–
Mindlin β_s (ϑ_r)	$\kappa_{ss} = \bar{b}_{r,s}$, $\gamma_{sz} = \bar{b}_r$	$\kappa_{xs} = \bar{b}_r$	–

2.2. Periodic boundary conditions

The cell is one period of an infinitely wide panel. The displacement field at the side edge $X = -W$ must therefore equal that at $X = 0$, for all components and at every x:

$$\mathcal{U}_\alpha(X = -W,\, x) = \mathcal{U}_\alpha(X = 0,\, x), \qquad \alpha = 1, \ldots, 6, \qquad 0 \le x \le L, \tag{13}$$

where $\mathcal{U}_\alpha$ are the six degrees of freedom of a shell node, three translations and three rotations. This is the condition that a shell finite element model imposes with one constraint equation per pair of side nodes and degree of freedom [5–7]. On the skin side edges, whose normal is Y, the degrees of freedom correspond to the GBT quantities in Table 2. Since all of them are linear in the amplitude functions, Eq. (13) holds for every x and every set of amplitude functions if, and only if, every mode satisfies it. The periodic condition is therefore built into the modes, family by family.

Table 2. Degrees of freedom of a skin side node and the corresponding conditions in GBT.

Shell DOF	Meaning at a skin side node	Periodic condition in GBT
$\mathcal{U}_1$	in-plane displacement U_X	$U_{X,s_L} = U_{X,s_R}$, Eq. (14)
$\mathcal{U}_2$	deflection U_Y	$U_{Y,s_L} = U_{Y,s_R}$, Eq. (14)
$\mathcal{U}_3$	axial displacement u	warping of G and D modes, Eq. (16); shear modes, Eq. (17); axial mode uniform
$\mathcal{U}_4$	rotation $\phi_x = -w_{,x} + \beta_x$	$w_{,x}$ periodic because U_Y is; $\Omega_{s_L} = \Omega_{s_R}$, Eq. (18)

Shell DOF	Meaning at a skin side node	Periodic condition in GBT
$\mathcal{U}_6$	rotation $\theta + \beta_s$	$\theta_{s_L} = \theta_{s_R}$, Eq. (14); $\bar{b}_{s_L} = \bar{b}_{s_R}$, Eq. (18)
$\mathcal{U}_5$	drilling rotation	not a degree of freedom of the plate theory

For the conventional modes, the conditions $U_{X,s_L} = U_{X,s_R}$, $U_{Y,s_L} = U_{Y,s_R}$ and $\theta_{s_L} = \theta_{s_R}$ are the rows of a constraint matrix acting on the stacked nodal vector,

$$\mathbf{R}\begin{bmatrix}\mathbf{U}\\ \Theta\end{bmatrix} = \mathbf{0}, \qquad \mathbf{R} = \begin{bmatrix}\mathbf{e}_{X,s_L}^{\mathsf{T}} - \mathbf{e}_{X,s_R}^{\mathsf{T}}\\ \mathbf{e}_{Y,s_L}^{\mathsf{T}} - \mathbf{e}_{Y,s_R}^{\mathsf{T}}\\ \mathbf{e}_{\theta,s_L}^{\mathsf{T}} - \mathbf{e}_{\theta,s_R}^{\mathsf{T}}\end{bmatrix} \in \mathbb{R}^{3\times 3N}, \tag{14}$$

where $\mathbf{e}_{X,q}$, $\mathbf{e}_{Y,q}$ and $\mathbf{e}_{\theta,q}$ extract $U_{X,q}$, $U_{Y,q}$ and θ_q. Let $\mathbf{V}_g \in \mathbb{R}^{3N\times n_g}$ hold the nodal fields $[\mathbf{U};\Theta]$ of the n_g modes of family $g \in \{G, D, L, T\}$ of the unconstrained cell. The constraint is imposed family by family, so that the mechanical meaning of the families is kept:

1. the columns of $\mathbf{V}_g$ that satisfy $\mathbf{R}\mathbf{v} = \mathbf{0}$ are kept unchanged;
2. the remaining columns are replaced by a basis of their constrained combinations, $\mathbf{V}_g \leftarrow \mathbf{V}_g\mathbf{Z}_g$ with $\mathbf{Z}_g = \ker(\mathbf{R}\mathbf{V}_g)$;
3. the vectors of $\ker\left(\mathbf{R}\left[\mathbf{V}_G\ \mathbf{V}_D\ \mathbf{V}_L\ \mathbf{V}_T\right]\right)$ that are not yet spanned (combinations of several families) are added, to the family T if they elongate a segment ($\mathbf{E}_\ell\mathbf{U} \neq \mathbf{0}$) and to the family D otherwise;
4. each column is scaled to a unit maximum nodal displacement or rotation.

The resulting modal matrix $\mathbf{V} = [\mathbf{V}_G\ \mathbf{V}_D\ \mathbf{V}_L\ \mathbf{V}_T]$ satisfies $\mathbf{R}\mathbf{V} = \mathbf{0}$ and spans the whole periodic subspace $\ker\mathbf{R}$. Consequently, for arbitrary amplitude functions,

$$\mathbf{R}\sum_i \begin{bmatrix}\mathbf{U}_i\\ \Theta_i\end{bmatrix}\varphi_i(x) = \mathbf{0} \qquad \text{for all } x, \tag{15}$$

and the same holds for the derivatives of the amplitude functions and for any subset of the modes. The rigid rotation of the cross-section about the x-axis moves the two side nodes by different amounts ($U_{Y,s_L} - U_{Y,s_R} = -W\omega$ for a rotation ω) and is eliminated: a torsional rigid-body motion of one cell is not compatible with an infinitely wide panel. The two rigid translations remain. On the other hand, the side nodes may move laterally and rotate, provided that they do so identically, which yields 13 distortional modes (Table 3).

For the warping of the G and D modes, the periodic condition $\bar{u}_{s_L} = \bar{u}_{s_R}$ is added to the constraint of Eq. (12). With the segment difference matrix $\mathbf{H}$, $(\mathbf{H}\bar{\mathbf{u}})_p = \bar{u}_{q'} - \bar{u}_q$, the weight matrix $\mathbf{W} = \mathrm{diag}(\hat{A}_{66,p}/b_p)$, the vector $\mathbf{g}$ of the values $b_p\,\bar{v}_p|_{\eta=1/2}$ and the constraint matrix $\mathbf{C}_u = \left[\mathbf{c}_{ax}^{\mathsf{T}};\ \mathbf{e}_{s_L}^{\mathsf{T}} - \mathbf{e}_{s_R}^{\mathsf{T}}\right]$, the warping of each mode follows from the Karush–Kuhn–Tucker system

$$\begin{bmatrix}\mathbf{H}^{\mathsf{T}}\mathbf{W}\mathbf{H} & \mathbf{C}_u^{\mathsf{T}}\\ \mathbf{C}_u & \mathbf{0}\end{bmatrix}\begin{bmatrix}\bar{\mathbf{u}}\\ \mu\end{bmatrix} = \begin{bmatrix}-\mathbf{H}^{\mathsf{T}}\mathbf{W}\mathbf{g}\\ \mathbf{0}\end{bmatrix}, \tag{16}$$

which is the stationarity condition of Eq. (12) with the Lagrange multipliers μ.

The shear modes are the nodal hat functions of the warping. The hat functions of the two side nodes are merged into one mode with a common warping value,

$$h_{s_R} \leftarrow h_{s_R} + h_{s_L}, \qquad h_{s_L} \text{ removed}, \tag{17}$$

and in the same way the nodal functions of the Mindlin modes of s_L are added to those of s_R,

$$\Omega_{s_R} \leftarrow \Omega_{s_R} + \Omega_{s_L}, \qquad \bar{b}_{s_R} \leftarrow \bar{b}_{s_R} + \bar{b}_{s_L}, \qquad \Omega_{s_L},\ \bar{b}_{s_L} \text{ removed}. \tag{18}$$

The uniform axial mode is periodic by construction. Equations (14)–(18) make every mode satisfy Eq. (13). Any linear combination with arbitrary amplitude functions, and in particular any truncated subset of the modes, therefore satisfies the periodic condition exactly, and no penalty terms or Lagrange multipliers appear in the member analysis. The resulting numbers of modes are listed in Table 3.

Table 3. Number of deformation modes per family of the periodic cell (the numbers do not depend on the lay-up).

b_s [mm]	G	D	L	T	S (incl. axial)	M	total
150	2	13	166	92	91	205	569
285.5	2	13	138	78	77	174	482

The periodic condition cannot be added afterwards to a set of classical modes. Classical GBT generates the conventional modes from a few natural nodes of an open contour. The periodic condition couples the two ends of the contour, so that the cell behaves like a closed loop with respect to its in-plane kinematics: the rigid rotation is no longer admissible, the warping must take the same value on both sides of the cut, and the classical hierarchy has to be recomputed. Imposed afterwards on a truncated set of classical modes, the three conditions of Eq. (14) remove three degrees of freedom from a small space and leave too few admissible modes. Here, the modes are generated from a complete nodal basis, and the constraint is applied before any truncation.

The periodic cell is also related to Bloch–Floquet theory. For an infinite panel with the stringer pitch b_s, the buckling modes are Bloch waves, $\mathbf{u}(X + b_s, x) = e^{\mathrm{i}\kappa b_s}\,\mathbf{u}(X, x)$ [12]. A periodic cell of width $2b_s$ admits exactly the two real cases $\kappa b_s = 0$, in which the mode repeats in every bay, and $\kappa b_s = \pi$, in which it alternates in sign from bay to bay.

For visualisation, the conventional modes can be ordered as in classical GBT by the generalised eigenvalue problem $\hat{\mathbf{B}}\,\psi = \Lambda\left(k_r^2\hat{\mathbf{D}}_1 + k_r^4\hat{\mathbf{C}}\right)\psi$ at a reference wavenumber k_r, with the modal matrices of Section 2.3; a small value of Λ indicates a preference for long waves. The first eight such modes of the periodic cell are shown in Figure 2. The two side nodes move identically in every mode, including the lateral rigid translation (mode 2), while the rigid rotation is absent.

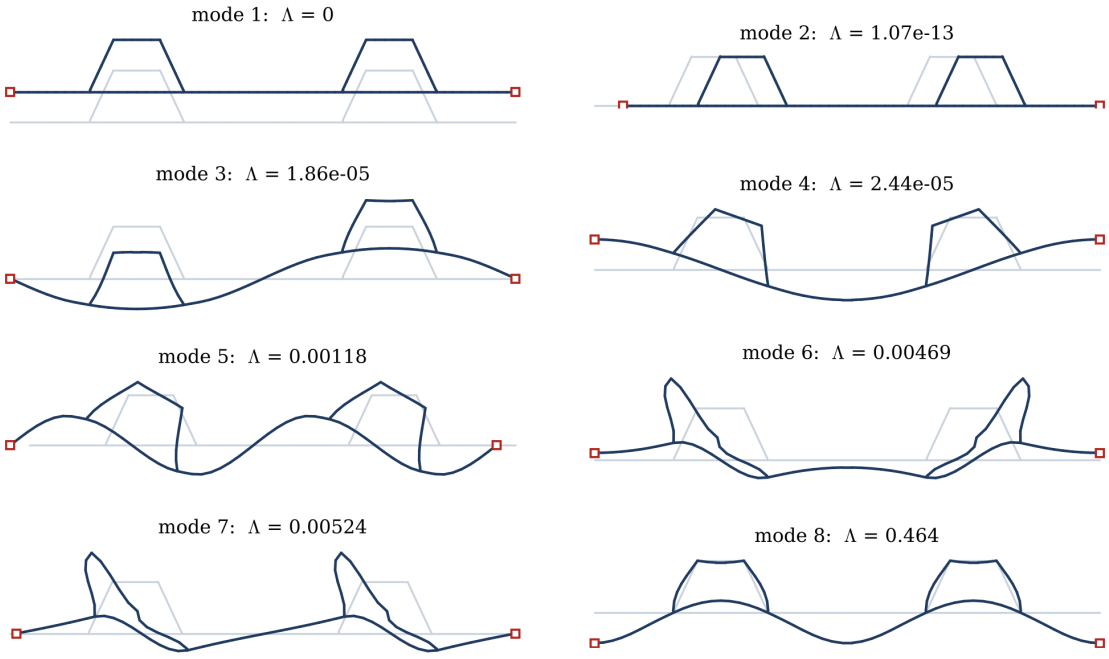


Figure 2. First eight hierarchical conventional modes of the periodic cell ($b_s = 150$ mm, Cross-ply lay-up, $k_r = 2\pi/b_s$; in-plane deformation, grey: undeformed). The side nodes s_L and s_R (squares) move identically in every mode.

2.3. Member analysis

The variation of the strain energy is expressed in terms of the amplitude functions and their variations,

$$\delta U = \int_L \int_b \int_t (\sigma_{xx}\delta\varepsilon_{xx} + \sigma_{ss}\delta\varepsilon_{ss} + \tau_{xs}\delta\gamma_{xs} + \tau_{xz}\delta\gamma_{xz} + \tau_{sz}\delta\gamma_{sz})\, dz\, ds\, dx = \int_0^L \delta\Phi^{\mathsf{T}}\, \Xi\, \Phi\, dx, \quad (19)$$

where the vector Φ collects the amplitude functions of all modes and their first and second derivatives,

$$\Phi = \begin{bmatrix} \varphi \\ \varphi^I \\ \varphi^{II} \end{bmatrix}, \qquad \varphi^I = \frac{d\varphi}{dx}, \qquad \varphi^{II} = \frac{d^2\varphi}{dx^2}, \quad (20)$$

with φ containing φ_i, φ_j, χ_r and ϑ_r. With the generalised strains of Table 1 written as $\hat{\varepsilon} = \sum_d \mathbf{E}^{(d)}(s)\, \varphi^{(d)}(x)$, $d \in \{0, I, II\}$, the GBT matrix Ξ consists of the blocks

$$\Xi = \begin{bmatrix} \Xi^{(0,0)} & \Xi^{(0,I)} & \Xi^{(0,II)} \\ \Xi^{(I,0)} & \Xi^{(I,I)} & \Xi^{(I,II)} \\ \Xi^{(II,0)} & \Xi^{(II,I)} & \Xi^{(II,II)} \end{bmatrix}, \qquad \Xi^{(d,e)}_{kl} = \int_\Gamma \mathbf{E}^{(d)\mathsf{T}}_k\, \mathbf{C}\, \mathbf{E}^{(e)}_l\, ds, \quad (21)$$

where Γ is the mid-line of the cross-section and $\Xi^{(e,d)} = \Xi^{(d,e)\mathsf{T}}$. As in the GBT–Ritz approach [49], Ξ is decomposed into the components of the conventional modes, $\hat{\Xi}$, of the shear modes, $\check{\Xi}$, and of the interaction between them, $\tilde{\Xi}$, supplemented here by the components Ξ^M that involve the Mindlin modes. For the conventional modes,

$$\hat{\Xi} = \begin{bmatrix} \hat{\mathbf{B}} & \hat{\mathbf{F}}_1 & \hat{\mathbf{D}}_2 \\ \hat{\mathbf{F}}_1^{\mathsf{T}} & \hat{\mathbf{D}}_1 & \hat{\mathbf{H}}_1 \\ \hat{\mathbf{D}}_2^{\mathsf{T}} & \hat{\mathbf{H}}_1^{\mathsf{T}} & \hat{\mathbf{C}} \end{bmatrix}, \quad (22)$$

and substituting Table 1 and Eq. (8) into Eq. (21) gives, for symmetric and unsymmetric laminates, with $\gamma_i = \bar{u}_{i,s} + \bar{v}_i$ and the conventional modes i and k,

$$\hat{B}_{ik} = \int_\Gamma \Big[A_{22}\, \bar{v}_{i,s}\bar{v}_{k,s} - B_{22}\left(\bar{v}_{i,s}\bar{w}_{k,ss} + \bar{w}_{i,ss}\bar{v}_{k,s}\right) + D_{22}\, \bar{w}_{i,ss}\bar{w}_{k,ss}\Big] ds, \quad (23)$$

$$\hat{F}_{1,ik} = \int_\Gamma \Big[A_{26}\, \bar{v}_{i,s}\gamma_k - B_{26}\left(2\bar{v}_{i,s}\bar{w}_{k,s} + \bar{w}_{i,ss}\gamma_k\right) + 2D_{26}\, \bar{w}_{i,ss}\bar{w}_{k,s}\Big] ds, \quad (24)$$

$$\hat{D}_{2,ik} = \int_\Gamma \Big[A_{12}\, \bar{v}_{i,s}\bar{u}_k - B_{12}\left(\bar{v}_{i,s}\bar{w}_k + \bar{w}_{i,ss}\bar{u}_k\right) + D_{12}\, \bar{w}_{i,ss}\bar{w}_k\Big] ds, \quad (25)$$

$$\hat{D}_{1,ik} = \int_\Gamma \Big[A_{66}\, \gamma_i\gamma_k - 2B_{66}\left(\gamma_i\bar{w}_{k,s} + \bar{w}_{i,s}\gamma_k\right) + 4D_{66}\, \bar{w}_{i,s}\bar{w}_{k,s}\Big] ds, \quad (26)$$

$$\hat{H}_{1,ik} = \int_\Gamma \Big[A_{16}\, \gamma_i\bar{u}_k - B_{16}\left(\gamma_i\bar{w}_k + 2\bar{w}_{i,s}\bar{u}_k\right) + 2D_{16}\, \bar{w}_{i,s}\bar{w}_k\Big] ds, \quad (27)$$

$$\hat{C}_{ik} = \int_\Gamma \Big[A_{11}\, \bar{u}_i\bar{u}_k - B_{11}\left(\bar{u}_i\bar{w}_k + \bar{w}_i\bar{u}_k\right) + D_{11}\, \bar{w}_i\bar{w}_k\Big] ds. \quad (28)$$

The transverse membrane stiffness enters through the A_{22} term of $\hat{\mathbf{B}}$, the bending–twisting coupling through $\hat{\mathbf{F}}_1$ and $\hat{\mathbf{H}}_1$, and the membrane–bending coupling through all B_{ij} terms. For the shear modes j and l,

$$\breve{\Xi} = \begin{bmatrix} \breve{\mathbf{D}}_1 & \breve{\mathbf{H}}_1 & \mathbf{0} \\ \breve{\mathbf{H}}_1^{\mathsf{T}} & \breve{\mathbf{C}} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix},$$

$$\breve{D}_{1,jl} = \int_\Gamma A_{66}\,\bar{u}_{j,s}\bar{u}_{l,s}\,ds, \quad \breve{H}_{1,jl} = \int_\Gamma A_{16}\,\bar{u}_{j,s}\bar{u}_l\,ds, \quad \breve{C}_{jl} = \int_\Gamma A_{11}\,\bar{u}_j\bar{u}_l\,ds, \tag{29}$$

and for the interaction between a conventional mode i (rows) and a shear mode j (columns),

$$\tilde{\Xi} = \begin{bmatrix} \tilde{\mathbf{F}} & \tilde{\mathbf{D}}_2 & \mathbf{0} \\ \tilde{\mathbf{D}}_1 & \tilde{\mathbf{H}}_1 & \mathbf{0} \\ \tilde{\mathbf{H}}_3 & \tilde{\mathbf{C}} & \mathbf{0} \end{bmatrix}, \tag{30}$$

$$\begin{aligned} \tilde{F}_{ij} &= \int_\Gamma \left(A_{26}\,\bar{v}_{i,s} - B_{26}\,\bar{w}_{i,ss}\right)\bar{u}_{j,s}\,ds, \qquad \tilde{D}_{2,ij} = \int_\Gamma \left(A_{12}\,\bar{v}_{i,s} - B_{12}\,\bar{w}_{i,ss}\right)\bar{u}_j\,ds, \\ \tilde{D}_{1,ij} &= \int_\Gamma \left(A_{66}\,\gamma_i - 2B_{66}\,\bar{w}_{i,s}\right)\bar{u}_{j,s}\,ds, \qquad \tilde{H}_{1,ij} = \int_\Gamma \left(A_{16}\,\gamma_i - 2B_{16}\,\bar{w}_{i,s}\right)\bar{u}_j\,ds, \\ \tilde{H}_{3,ij} &= \int_\Gamma \left(A_{16}\,\bar{u}_i - B_{16}\,\bar{w}_i\right)\bar{u}_{j,s}\,ds, \qquad \tilde{C}_{ij} = \int_\Gamma \left(A_{11}\,\bar{u}_i - B_{11}\,\bar{w}_i\right)\bar{u}_j\,ds. \end{aligned} \tag{31}$$

The components Ξ^M of the Mindlin modes follow from Eq. (21) and Table 1 in the same way; they contain the transverse shear stiffness, e.g. $\int_\Gamma A_{s,11}\,\bar{c}_r\bar{c}_{r'}\,ds$ in the block $(0,0)$ of two β_x modes, and the bending stiffnesses **D** and **B** through the curvatures of Table 1. All blocks are evaluated by four-point Gauss quadrature on every segment.

The panel is loaded at both loaded edges by a compressive line load $q(s)$ per unit length, which acts at the distance $e_q(s)$ from the wall reference surface. Its external work is

$$W_{ext} = \int_\Gamma q(s)\,\left[u_q(0,s) - u_q(L,s)\right]\,ds, \qquad u_q = u - e_q\,w_{,x}, \tag{32}$$

where u_q is the axial displacement of the fibre at which the load acts. The pre-buckling state is not assumed but computed: it is the first-order solution of the stiffness relation under this load and under the end and side conditions of the cell, $\mathbf{K}\mathbf{a}^0 = \mathbf{f}$, with **K** and **a** as defined below and the load vector **f** from the variation of Eq. (32). The membrane forces of this state are

$$[S_{xx},\ S_{ss},\ S_{xs}]^{\mathsf{T}} = \mathbf{A}\,\varepsilon^0 + \mathbf{B}\,\kappa^0, \tag{33}$$

with the membrane strains ε^0 and curvatures κ^0 of the pre-buckling state. They vary along s and x: the load is introduced at the ends and shared between skin and stringer by shear lag, the eccentric foot load bends the cell near the ends, and the restrained Poisson expansion produces transverse membrane forces S_{ss}.

The loading is characterised by the second-order energy of the membrane forces,

$$V = \frac{1}{2}\int_0^L \int_\Gamma \sum_{c\in\{u,v,w\}} \Big[S_{xx}\,c_{,x}^2 + S_{ss}\,c_{,s}^2 + 2S_{xs}\,c_{,x}\,c_{,s}\Big]\,ds\,dx = \frac{1}{2}\int_0^L \Phi^{\mathsf{T}}\mathcal{X}(x)\,\Phi\,dx, \tag{34}$$

in which all three displacement components and all three membrane forces are kept. For a conventional mode, the displacement gradients are $u_{,x} = \bar{u}_i\varphi_i^{II}$, $u_{,s} = \bar{u}_{i,s}\varphi_i^{I}$, $v_{,x} = \bar{v}_i\varphi_i^{I}$, $v_{,s} = \bar{v}_{i,s}\varphi_i$, $w_{,x} = \bar{w}_i\varphi_i^{I}$ and $w_{,s} = \bar{w}_{i,s}\varphi_i$, and for a shear mode $u_{,x} = \bar{u}_j\varphi_j^{I}$ and $u_{,s} = \bar{u}_{j,s}\varphi_j$; the Mindlin modes do not move the mid-surface and do not contribute to $\mathcal{X}$. Since the membrane forces vary along x, the matrix $\mathcal{X}(x)$ is evaluated at every integration point along the cell. For the conventional modes and a uniform membrane force S_{xx} with $S_{ss} = S_{xs} = 0$, Eq. (34) reduces to the geometric matrix of the GBT–Ritz approach [46,49],

$$\mathcal{X} = \begin{bmatrix} \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathcal{X}^{VW} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathcal{X}^{U} \end{bmatrix}, \qquad \mathcal{X}^{U}_{ik} = \int_\Gamma S_{xx}\, \bar{u}_i \bar{u}_k \, ds, \qquad \mathcal{X}^{VW}_{ik} = \int_\Gamma S_{xx}\, (\bar{v}_i \bar{v}_k + \bar{w}_i \bar{w}_k)\, ds. \tag{35}$$

Following the Ritz method, Φ is written as the product of the matrix Ψ of shape functions and the vector $\mathbf{a}$ of unknown coefficients,

$$\Phi = \Psi\, \mathbf{a}, \qquad \varphi(x) = \sum_{b=1}^{n_b} N_b(x)\, a_b. \tag{36}$$

In the GBT–Ritz approach for members, trigonometric series are used [49]. In the panel cell, the pre-buckling state varies strongly near the loaded edges, where the load enters the stringer, and a single trigonometric series represents this variation only with many terms. The shape functions N_b are therefore cubic B-splines on an open, uniform knot vector with spans of length $h_x = \min(15\text{ mm},\ \max(2\text{ mm},\ W/40),\ L/10)$, i.e. $h_x = b_s/20$ for all panels studied. They are local, represent simply supported and clamped edges alike, and with the open knot vector $\varphi(0) = a_1$, $\varphi(L) = a_{n_b}$ and $\varphi^I(0) = 0 \Leftrightarrow a_2 = a_1$, so that the essential end conditions act directly on the end coefficients. Two end conditions of the loaded edges are considered, as in the shell models:

- **Simply supported (SS):** the out-of-plane displacement of every end edge vanishes, $U_Y = 0$ at every node at $x = 0$ and $x = L$, i.e. $\sum_i U_{Y,qi}\, \varphi_i(0) = \sum_i U_{Y,qi}\, \varphi_i(L) = 0$ for every node q. These linear relations between the end coefficients are imposed by a null-space reduction; the in-plane displacement U_X, the warping and the rotations of the end cross-sections remain free. Rigid-body motions are removed by fixing the axial mode and U_X at the side node s_R at $x = 0$.
- **Clamped (CC):** $\varphi_i = \varphi_i^I = 0$ at both ends for every conventional mode (in-plane displacements and the rotations $w_{,x}$ and θ vanish), $\chi_r = \vartheta_r = 0$ at both ends, and $\varphi_j(0) = 0$ for the shear modes, so that $u = 0$ on the whole edge $x = 0$; the edge $x = L$ remains free to warp under the uniform load.

In classical GBT, simply supported ends are represented by $\varphi_i = \sin(h\pi x/L)$, $h = 1, 2, ...$, which suppresses all in-plane displacements of the end cross-sections while leaving their warping free. The present SS condition is weaker: like the shell model, it only suppresses the out-of-plane displacement U_Y of the end edges.

With Eq. (36), the stationarity of the total potential energy, $\delta(U + \lambda V) = 0$, leads to the eigenvalue problem

$$(\mathbf{K} + \lambda\, \mathbf{G})\, \mathbf{a} = \mathbf{0}, \tag{37}$$

where $\mathbf{K}$ is the stiffness matrix associated with Ξ and $\mathbf{G}$ the geometric stiffness matrix arising from the membrane forces $\mathcal{X}$ of the reference load,

$$\mathbf{K} = \int_0^L \Psi^{\mathsf{T}} \left(\hat{\Xi} + \check{\Xi} + \tilde{\Xi} + \Xi^M \right) \Psi \, dx, \qquad \mathbf{G} = \int_0^L \Psi^{\mathsf{T}} \mathcal{X}\, \Psi \, dx, \tag{38}$$

in which every component of Ξ is placed at the rows and columns of its modes, and $\tilde{\Xi}$ also at their transposed position. The integrals are evaluated with four Gauss points per spline span. The smallest positive eigenvalue λ multiplies the reference load of 1 N/mm, so that the critical line load is $N_{cr} = \lambda \times 1$ N/mm, and the eigenvector $\mathbf{a}$ defines the buckling mode. Since $\mathbf{G}$ is indefinite (the pre-buckling state contains membrane shear and some transverse tension), Eq. (37) is solved for the largest values of $\mu = 1/\lambda$ of $-\mathbf{G}\,\mathbf{a} = \mu\, \mathbf{K}\,\mathbf{a}$ by the implicitly restarted Lanczos method [54] with a sparse factorisation of $\mathbf{K}$. For $b_s = 150$ mm and $L = 225$ mm the model has $569 \times 33 = 18\,777$ coefficients before the end conditions are applied.

Two numerical aspects are noted. First, at some intermediate nodes of the inclined webs the two segment tangents are identical up to round-off, and the orthonormal basis of the tangents contains a spurious second Mindlin vector normal to the web. For it $\bar{c}_r = \mathbf{t} \cdot \Omega \approx 10^{-15}$ on every segment, so that the mode carries neither strain energy nor displacement gradients (11 such modes for $b_s = 150$ mm and 8 for $b_s = 285.5$ mm). Such modes cannot change any eigenvalue, but they make $\mathbf{K}$ singular; they are identified by their vanishing diagonal of Ξ and their coefficients are held. Second, at a cross-section integration point only a few modes

are non-zero (about 25 of 569 for $b_s = 150$ mm), so that the cross-section factors of $\mathcal{X}$ are stored as sparse matrices that are computed once per cross-section; for every panel length only the membrane forces change.

The formulation contains classical GBT as a special case. For the conventional modes alone, walls without coupling ($\mathbf{B} = \mathbf{0}$, $A_{16} = A_{26} = D_{16} = D_{26} = 0$), and a uniform membrane force S_{xx}, the matrices $\hat{\mathbf{F}}_1$ and $\hat{\mathbf{H}}_1$ vanish, and the Euler–Lagrange equations of $\delta(U + \lambda V) = 0$ with Eqs. (22) and (35) are

$$\hat{\mathbf{C}}\,\varphi^{IIII} - \hat{\mathbf{D}}\,\varphi^{II} + \hat{\mathbf{B}}\,\varphi + \lambda\left(\mathcal{X}^{U}\varphi^{IIII} - \mathcal{X}^{VW}\varphi^{II}\right) = \mathbf{0}, \qquad \hat{\mathbf{D}} = \hat{\mathbf{D}}_1 - \hat{\mathbf{D}}_2 - \hat{\mathbf{D}}_2^{\mathsf{T}}, \tag{39}$$

where the integration by parts of the $\hat{\mathbf{D}}_2$ terms gives the classical matrix $\hat{\mathbf{D}}$. Without the warping term $\mathcal{X}^{U}$, Eq. (39) is the classical GBT buckling equation of Schardt [24] and of Silvestre and Camotim for orthotropic members [30]. For simply supported ends and $\varphi = \hat{\varphi}\sin k_h x$, $k_h = h\pi/L$, it becomes the algebraic eigenvalue problem

$$\left[\hat{\mathbf{B}} + k_h^2\,\hat{\mathbf{D}} + k_h^4\,\hat{\mathbf{C}} + \lambda\left(k_h^2\,\mathcal{X}^{VW} + k_h^4\,\mathcal{X}^{U}\right)\right]\hat{\varphi} = \mathbf{0}, \tag{40}$$

whose smallest eigenvalue over h is the classical GBT signature curve. The periodic condition does not alter this structure: it only restricts the admissible modes of the cross-section. The present model differs from the classical one only by the terms that the panel requires: laminate coupling ($\hat{\mathbf{F}}_1$, $\hat{\mathbf{H}}_1$, B_{ij}), shear and Mindlin modes, a non-uniform pre-buckling state with S_{ss} and S_{xs}, and end conditions that support the loaded edges without restraining their warping.

2.4. Mode decomposition

The eigenvector $\mathbf{a}$ defines the buckling mode in terms of the GBT modes but does not directly show the contribution of the individual deformation mechanisms. As in the GBT–Ritz approach [49], an energy-based mode decomposition is therefore performed. The modes are grouped into the families $g \in \{G, D, L, T, S, M\}$, and $\mathbf{a}_g$ denotes the vector $\mathbf{a}$ with all coefficients set to zero except those of family g. The strain energy of the buckling mode, $U = \frac{1}{2}\int_0^L \Phi^{\mathsf{T}}\Xi\,\Phi\,dx = \frac{1}{2}\mathbf{a}^{\mathsf{T}}\mathbf{K}\,\mathbf{a}$, is divided into the contributions of the families,

$$U_g = \frac{1}{2}\,\mathbf{a}_g^{\mathsf{T}}\mathbf{K}\,\mathbf{a}, \qquad \rho_g = \frac{U_g}{U}, \qquad \sum_g \rho_g = 1. \tag{41}$$

If the families are orthogonal with respect to $\mathbf{K}$, U_g equals the energy of family g alone, $\frac{1}{2}\mathbf{a}_g^{\mathsf{T}}\mathbf{K}\,\mathbf{a}_g$, which is the form used for members [49]. In the panel cell, the distortional and local modes are not orthogonal: the D modes move the wall junctions without rotating the nodes, and the rotation modes of family L restore the continuity of the walls. The energies of the families alone then overestimate the families that are coupled to others and do not add up to U, whereas Eq. (41) divides the strain energy exactly. For instance, for the Unsym panel with simply supported loaded edges, $b_s = 150$ mm and $\beta = 1.5$, the energies of the families alone would assign 41 % to the distortional and 58 % to the local family, whereas Eq. (41) gives 1 % and 97 %.

3. Results and discussion

This section verifies the formulation and discusses the buckling behaviour of omega-stringer-stiffened panels with periodic side edges. The critical loads predicted by the GBT–Ritz formulation are compared with Abaqus shell models with periodic constraint equations for simply supported loaded edges and with an independent spline finite strip model for clamped loaded edges. Particular attention is given to the critical loads, the buckling modes, the contribution of the individual deformation modes and the influence of the laminate coupling. The geometry of the cross-section and the material properties are summarised in Table 4, and the lay-ups in Table 5. They include an isotropic reference, a cross-ply laminate without coupling, two symmetric angle-ply laminates with bending–twisting coupling ($D_{16}, D_{26} \neq 0$; Sym2 and the seven-ply stringer of Sym1 also with $A_{16}, A_{26} \neq 0$), and an unsymmetric laminate with membrane–bending coupling ($\mathbf{B} \neq \mathbf{0}$). Ply angles are measured from the stringer axis x. Two stringer pitches, $b_s = 150$ mm and 285.5 mm, and aspect ratios $0.5 \leq \beta \leq 3$ are considered.

Table 4. Geometry of the cross-section (Figure 1) and material properties.

Parameter	Value
stringer pitch b_s	150 mm, 285.5 mm
web root half-width b_3, web height h, web inclination	27.75 mm, 30 mm, 25° from the vertical
crown half-width $h_t = b_3 - h\tan 25^\circ$	13.76 mm
foot width $b_1 = 0.1\,(b_s - 2b_3)$	9.45 mm, 23.00 mm
skin and stringer thickness t_s, t_f	1.0 mm, 0.875 mm (8 and 7 plies of 0.125 mm)
stringer offset $c = (t_s + t_f)/2$	0.9375 mm
CFRP ply	$E_1 = 150$ GPa, $E_2 = 9$ GPa, $G_{12} = G_{13} = G_{23} = 5.3$ GPa, $\nu_{12} = 0.33$
isotropic material	$E = 200$ GPa, $\nu = 0.3$

Table 5. Lay-ups of skin and stringer (angles in degrees from the x-axis; subscript s: symmetric, subscript 2: repeated twice).

Name	Skin	Stringer	Coupling
Isotropic	–	–	none
Cross-ply	$[0/90/0/90]_s$	$[0/90/0/90/0/90/0]$	none
Sym1	$[45/-45/-45/45]_s$	$[45/-45/45/-45/45/-45/45]$	D_{16}, D_{26} (stringer also A_{16}, A_{26})
Sym2	$[60/-30/-30/60]_s$	$[60/-30/60/-30/60/-30/60]$	$A_{16}, A_{26}, D_{16}, D_{26}$
Unsym	$[0/30/60/90]_2$	$[0/30/60/90/0/30/60]$	**A**, **B**, **D** fully populated

A uniform compressive line load of 1 N/mm acts at both loaded edges on the skin, the webs and the crowns, and 2 N/mm on a foot, which carries skin and flange; the foot load acts at the height of its resultant, $Y = c/2$. The eigenvalue λ is therefore the critical line load N_{cr} in N/mm, the quantity reported by the buckling analysis of the shell model.

The shell models are linear buckling analyses in Abaqus with S4R and S3 elements and a seed of 3.7 mm. The stringer is tied to the skin by *TIE with the true offset of the mid-surfaces, and the loads are applied as shell edge loads. The simply supported loaded edges are modelled by $\mathcal{U}_2 = 0$ on every end edge of skin and stringer. For the periodic side edges, every skin node of the edge $X = -W$ is paired with the node of $X = 0$ at the same x, and one constraint equation $\mathcal{U}_\alpha(-W, x) - \mathcal{U}_\alpha(0, x) = 0$ is written for each pair and each of the six degrees of freedom, as in Eq. (13); at the two end corners the equation for $\mathcal{U}_2$ is omitted because $\mathcal{U}_2 = 0$ is already prescribed there. The rigid-body motions left by periodicity are removed by $\mathcal{U}_1 = \mathcal{U}_3 = 0$ at the corner $X = 0$, $x = 0$. The GBT model imposes the same physical conditions (Table 2). Abaqus results with periodic side edges are available for the simply supported loaded edges.

For the clamped loaded edges, an in-house spline finite strip model (FSM) [10,11] serves as the reference. It discretises the same geometry, laminates, loads and end conditions with flat laminated Mindlin strips of width $b_s/50$ (twice as fine as the GBT segments), linear interpolation across each strip, a mid-strip transverse shear strain against shear locking, and six degrees of freedom per nodal line in global axes. Along x it uses the same cubic B-splines as the GBT model. The web roots are joined to the skin by rigid offset links, the direct analogue of the *TIE of the shell model, and the periodic condition is applied node by node by merging the six degrees of freedom of the two side nodal lines for every spline coefficient, which is literally the constraint set of the shell model. The pre-buckling state and the geometric stiffness are computed as in Section 2.3. The FSM shares the plate theory with GBT but none of the cross-section discretisation: it has no deformation modes, a finer mesh, rigid links instead of link walls, and boundary conditions on nodal degrees of freedom instead of built into modes. Its agreement with GBT is therefore an independent check of the mode construction. With simply supported loaded edges it agrees with Abaqus within 0.59 % on average and 1.37 % at most (Table 6).

3.1. Verification of the formulation

Since the periodic condition is built into the modes, every buckling mode satisfies it identically. For the cross-ply cell with $b_s = 150$ mm, the largest difference between the two side nodes over all conventional modes is

7.9×10^{-15} for the in-plane displacements, 1.1×10^{-16} for the rotation θ and 2.3×10^{-11} for the warping (modes scaled to a unit maximum), and it is zero for the shear and Mindlin modes. In the buckling mode of the Unsym panel with clamped loaded edges, $b_s = 150$ mm and $\beta = 1.5$, the largest differences between the side edges along the whole length are 1.2×10^{-18} (U_X), 6.9×10^{-18} (U_Y), 1.4×10^{-20} (θ) and 1.1×10^{-15} (warping), with modal amplitudes of order 10^{-5} to 10^{-1}, while the side edges themselves move.

The explicit modal matrices of Eqs. (23)–(31) were compared with the general expression of Eq. (21) evaluated by the implementation for the Cross-ply, Sym2 and Unsym cells; all blocks agree to within 2×10^{-15} relative to their largest entry, i.e. to round-off. The classical limit of Section 2.3 was checked for the Cross-ply cell with $b_s = 150$ mm by restricting the model to the conventional modes and imposing a uniform compressive membrane force and the classical end condition $\varphi_i(0) = \varphi_i(L) = 0$. The B-spline Ritz solution then reproduces the minimum of the signature-curve eigenvalue problem of Eq. (40), 33.747 N/mm with three half-waves for $L = 225$ mm, to within 0.001 %; the same value is obtained for $L = 450$ mm with six half-waves.

3.2. Simply supported loaded edges

Figure 3 shows the critical load as a function of the aspect ratio for the two stringer pitches and all lay-ups, together with the Abaqus results, and Table 6 lists all 30 cases that were computed with Abaqus. GBT agrees with Abaqus within 0.57 % on average and 1.25 % at most; the FSM agrees within 0.59 % and 1.37 %, and GBT and FSM differ by 0.22 % on average. The accuracy is the same for the uncoupled and the coupled laminates (mean deviation 0.49–0.73 % for every lay-up), which confirms the treatment of the bending–twisting and membrane–bending coupling in Eqs. (23)–(31).

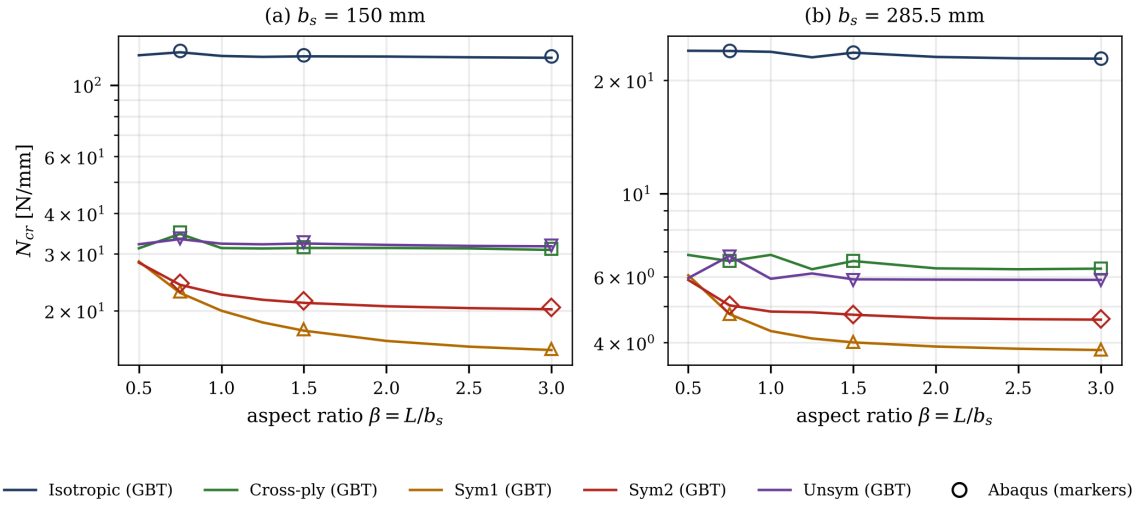


Figure 3. Critical load N_{cr} as a function of the aspect ratio β of the periodic cell with simply supported loaded edges for (a) $b_s = 150$ mm and (b) $b_s = 285.5$ mm: GBT (lines) and Abaqus shell models with periodic constraint equations (markers).

For the isotropic, cross-ply and unsymmetric cells, the critical load varies only little with the aspect ratio, with local maxima where the length of the cell does not accommodate an integer number of buckle half-waves of the preferred length, as known from the garland curves of plates. For $b_s = 150$ mm such a maximum occurs at $\beta = 0.75$, and the Abaqus results follow it (cross-ply: 34.848 N/mm in Abaqus and 34.530 N/mm in GBT at $\beta = 0.75$, against 31.464 and 31.243 N/mm at $\beta = 1.5$). For the angle-ply cells Sym1 and Sym2, the critical load decreases monotonically with β towards the value of the long cell. Their bending–twisting coupling skews the buckles, which therefore cannot adapt to the straight loaded edges by an integer number of half-waves, and the end conditions stiffen short cells.

Table 6. Critical loads N_{cr} [N/mm] with periodic side edges and simply supported loaded edges: Abaqus, GBT and spline finite strip model (FSM), and their deviations e from Abaqus.

Lay-up	b_s [mm]	β	Abaqus	GBT	e_{GBT} [%]	FSM	e_{FSM} [%]
Isotropic	150	0.75	127.650	126.284	−1.07	126.345	−1.02

Lay-up	b_s [mm]	β	Abaqus	GBT	e_{GBT} [%]	FSM	e_{FSM} [%]
	150	1.5	123.880	122.674	−0.97	122.617	−1.02
	150	3	122.740	121.412	−1.08	121.395	−1.10
	285.5	0.75	23.979	23.954	−0.10	24.022	+0.18
	285.5	1.5	23.720	23.707	−0.05	23.742	+0.09
	285.5	3	22.873	22.852	−0.09	22.898	+0.11
Cross-ply	150	0.75	34.848	34.530	−0.91	34.606	−0.69
	150	1.5	31.464	31.243	−0.70	31.258	−0.65
	150	3	31.061	30.810	−0.81	30.846	−0.69
	285.5	0.75	6.607	6.595	−0.18	6.618	+0.17
	285.5	1.5	6.609	6.597	−0.18	6.620	+0.17
	285.5	3	6.304	6.296	−0.13	6.310	+0.10
Sym1	150	0.75	22.880	22.675	−0.90	22.583	−1.30
	150	1.5	17.475	17.315	−0.92	17.272	−1.16
	150	3	15.202	15.078	−0.82	15.087	−0.76
	285.5	0.75	4.760	4.762	+0.04	4.772	+0.25
	285.5	1.5	4.011	4.004	−0.17	4.020	+0.22
	285.5	3	3.832	3.819	−0.34	3.840	+0.21
Sym2	150	0.75	24.291	24.024	−1.10	23.988	−1.25
	150	1.5	21.386	21.118	−1.25	21.144	−1.13
	150	3	20.415	20.174	−1.18	20.229	−0.91
	285.5	0.75	5.033	5.026	−0.14	5.041	+0.16
	285.5	1.5	4.758	4.744	−0.29	4.762	+0.08
	285.5	3	4.620	4.602	−0.39	4.623	+0.06
Unsym	150	0.75	33.489	33.277	−0.63	33.159	−0.99
	150	1.5	32.561	32.260	−0.92	32.115	−1.37
	150	3	31.825	31.616	−0.66	31.511	−0.99
	285.5	0.75	6.775	6.806	+0.46	6.805	+0.44
	285.5	1.5	5.880	5.896	+0.27	5.896	+0.27
	285.5	3	5.864	5.877	+0.22	5.877	+0.22

The deviations depend mainly on the stringer pitch. For $b_s = 285.5$ mm the mean deviation of GBT is 0.20 % (maximum 0.46 %). For $b_s = 150$ mm it is 0.93 % (maximum 1.25 %), and all 15 GBT results and all 15 FSM results lie below Abaqus (GBT −1.25 % to −0.63 %). This offset is therefore common to the two plate models and not caused by the GBT mode basis, and the refinement study of Section 3.5 shows that it is not a discretisation error of GBT either, since finer segments lower the GBT result slightly. Its likely origins are differences between the S4R shell elements and the plate model in the short skin bays of the narrow pitch, and the representation of the tie; it has not been resolved in this study.

3.3. Clamped loaded edges

The critical loads with clamped loaded edges are shown in Figure 4 and listed in Table 7. Clamping raises the critical load of short cells considerably, e.g. by 59 % for the isotropic cell with $b_s = 150$ mm and $\beta = 0.5$, and the curves decrease monotonically with β. For long cells the clamped values approach, and for some lay-ups fall slightly below, the simply supported ones (e.g. 118.13 against 121.41 N/mm for the isotropic cell with $\beta = 3$); the FSM gives the same trend (118.17 against 121.40 N/mm). In the present models the clamped edges also restrain the in-plane displacements of the loaded edges, which changes the pre-buckling state near the ends. GBT and FSM agree within 0.30 % on average and 1.17 % at most over the 30 cases, with the largest deviations for the Sym1 lay-up with $b_s = 285.5$ mm, where the long skin bays and the bending–twisting coupling produce steep gradients of the mode near the clamped ends (Section 3.5). Abaqus results with periodic side edges and clamped loaded edges are not yet available, so that the clamped results are verified by the FSM only.

Table 7. Critical loads N_{cr} [N/mm] with periodic side edges and clamped loaded edges: spline finite strip model (FSM) and GBT, and the deviation e of GBT from the FSM.

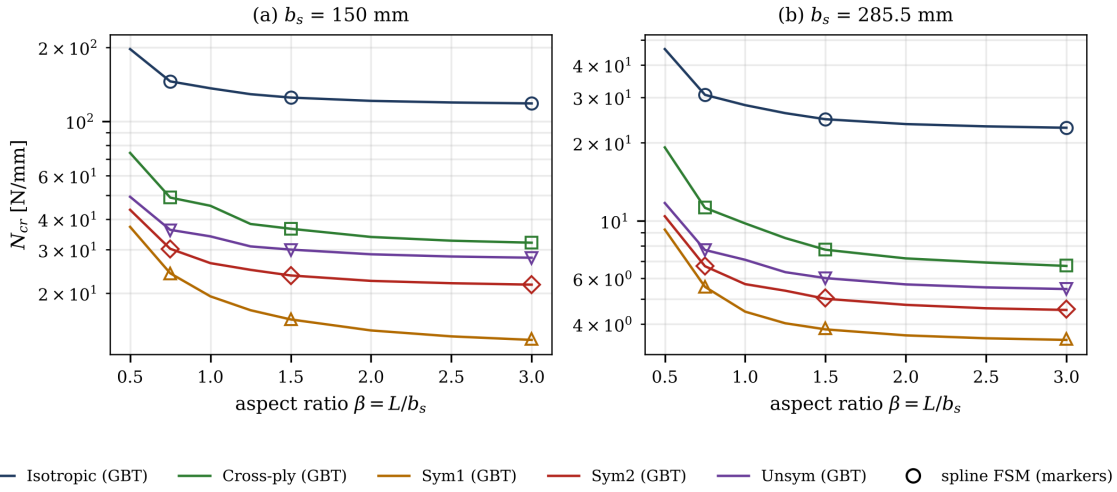


Figure 4. Critical load N_{cr} as a function of the aspect ratio β of the periodic cell with clamped loaded edges for (a) $b_s = 150$ mm and (b) $b_s = 285.5$ mm: GBT (lines) and spline finite strip model (markers).

Lay-up	b_s [mm]	β	FSM	GBT	e_{GBT} [%]
Isotropic	150	0.75	145.165	145.225	+0.04
	150	1.5	124.834	124.801	−0.03
	150	3	118.166	118.133	−0.03
	285.5	0.75	30.694	30.645	−0.16
	285.5	1.5	24.736	24.675	−0.25
	285.5	3	22.935	22.870	−0.28
Cross-ply	150	0.75	48.899	48.878	−0.04
	150	1.5	36.538	36.503	−0.10
	150	3	32.052	32.027	−0.08
	285.5	0.75	11.243	11.228	−0.13
	285.5	1.5	7.754	7.737	−0.22
	285.5	3	6.716	6.699	−0.25
Sym1	150	0.75	23.959	24.047	+0.37
	150	1.5	15.654	15.606	−0.31
	150	3	12.984	12.882	−0.79
	285.5	0.75	5.580	5.564	−0.29
	285.5	1.5	3.858	3.813	−1.17
	285.5	3	3.510	3.471	−1.11
Sym2	150	0.75	30.135	30.252	+0.39
	150	1.5	23.579	23.563	−0.07
	150	3	21.668	21.619	−0.23
	285.5	0.75	6.688	6.683	−0.07
	285.5	1.5	5.030	5.005	−0.50
	285.5	3	4.567	4.530	−0.81
Unsym	150	0.75	35.997	36.129	+0.37
	150	1.5	29.924	30.029	+0.35
	150	3	27.722	27.821	+0.36
	285.5	0.75	7.717	7.718	+0.01
	285.5	1.5	6.014	6.013	−0.02
	285.5	3	5.458	5.455	−0.05

3.4. Mode decomposition and buckling modes

Figure 5 shows the skin deflection of the critical mode of the Unsym cell with $b_s = 150$ mm and $\beta = 1.5$ for both end conditions. In both cases the mode is a local buckling of the skin bays between the stringers; the buckles are skewed by the laminate coupling, and the deflection at the side edges $X = -W$ and $X = 0$ is identical (largest difference 2.2×10^{-16} of the maximum deflection), so that the half-bays at the two sides join into one full bay of the same shape as the central one. With clamped loaded edges the buckles are concentrated away from the loaded edges.

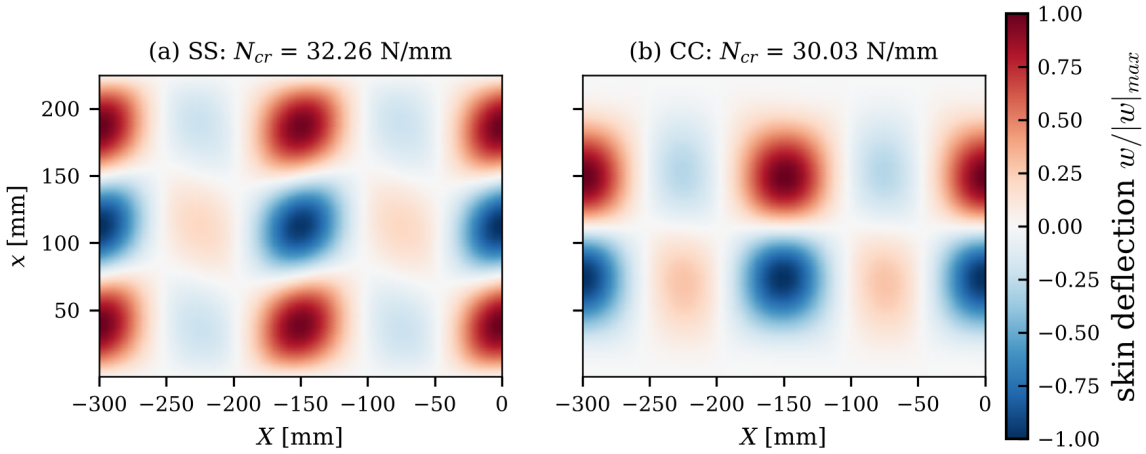


Figure 5. Skin deflection $w/|w|_{max}$ of the critical mode of the Unsym cell with periodic side edges, $b_s = 150$ mm and $\beta = 1.5$: (a) simply supported loaded edges; (b) clamped loaded edges. The stringers lie at $X = -75$ mm and $X = -225$ mm.

The energy-based mode decomposition of Eq. (41) quantifies these observations for all panels (Table 8). In every case the local family carries 96–101 % of the strain energy of the critical mode, the global family up to 3.5 % and the distortional family between −1.6 % and 1.1 %; the transverse-extension, shear and Mindlin families contribute less than 0.1 %. Contributions slightly above 100 % or below zero are possible in Eq. (41), because U_g contains the interaction energy of family g with the other families; small negative distortional contributions of the angle-ply laminates indicate that the distortional modes interact with the local skin buckles. The global participation is largest for the cross-ply cell with $b_s = 285.5$ mm, i.e. the local buckles of the wide skin bays are accompanied by a small overall bending of the cell. The decomposition does not change significantly with the aspect ratio, i.e. no transition to a distortional or global buckling mode occurs in the range studied.

Table 8. Energy-based participation ρ_g [%] of the mode families in the critical mode, range over $0.5 \leq \beta \leq 3$ (eight aspect ratios).

Lay-up	Ends	local (L)	distortional (D)	global (G)	T, S, M
Isotropic	SS	97.6 to 98.8	−0.2 to 0.1	1.4 to 2.4	< 0.04
	CC	97.8 to 99.0	−0.3 to 0.0	1.2 to 2.3	< 0.01
Cross-ply	SS	96.1 to 96.8	0.4 to 1.1	2.1 to 3.5	< 0.01
	CC	96.3 to 97.3	0.3 to 1.0	1.8 to 3.3	< 0.01
Sym1	SS	98.8 to 101.2	−1.5 to −0.5	0.2 to 1.7	< 0.05
	CC	99.1 to 101.3	−1.5 to −0.4	0.1 to 1.3	< 0.01
Sym2	SS	98.5 to 101.2	−1.6 to −0.4	0.3 to 1.9	< 0.04
	CC	98.5 to 100.4	−0.7 to −0.2	0.0 to 1.7	< 0.01
Unsym	SS	96.1 to 99.1	0.4 to 1.0	0.0 to 3.4	< 0.09
	CC	97.3 to 98.4	−0.1 to 0.3	1.5 to 2.7	< 0.02

3.5. Accuracy and efficiency

The default discretisation (segments of at most $b_s/25$, spline spans $h_x = b_s/20$) was checked by refining each parameter in turn (Table 9). Except for one case, all changes are within 0.50 %, i.e. below the deviation

from the reference models. The exception is the Sym1 cell with clamped loaded edges, $b_s = 285.5$ mm and $\beta = 1.5$: halving the spans repeatedly ($h_x/2$, $h_x/4$, $h_x/8$) raises the critical load by +0.26, +0.47 and +0.62 %, i.e. slowly, so that the converged value is estimated to lie about 1 % above the default result. The FSM, which uses the same spline spans, gives 3.858 N/mm for this case, i.e. the largest deviation between GBT and FSM in Table 7 stems mainly from the spline discretisation of GBT near the clamped ends of the wide skin bays with bending–twisting coupling. Finer segments lower the critical load slightly for $b_s = 150$ mm and therefore do not remove the offset against Abaqus noted in Section 3.2. The link walls were checked in the same way: for the isotropic cell with simply supported loaded edges, $b_s = 150$ mm and $\beta = 1.5$, the critical load is 122.667, 122.674 and 122.671 N/mm for $k_L = 10^2$, 10^3 and 10^4, whereas a link wall with the stiff twisting term $D_{66}^L = k_L D_{0,66}$ gives 150.620 N/mm (+22.78 %).

Table 9. Critical load N_{cr} [N/mm] with the default discretisation and with refined spline spans or cross-section segments (periodic side edges); in brackets the change from the default.

Case	Default	Spans $h_x/2$	Segments $b_s/40$
Isotropic, SS, $b_s = 150$, $\beta = 1.5$	122.6740	122.5921 (−0.07 %)	122.4080 (−0.22 %)
Sym2, SS, $b_s = 150$, $\beta = 3$	20.1744	20.1776 (+0.02 %)	20.1476 (−0.13 %)
Unsym, CC, $b_s = 150$, $\beta = 1.5$	30.0295	30.0275 (−0.01 %)	29.8795 (−0.50 %)
Sym1, CC, $b_s = 285.5$, $\beta = 3$	3.4715	3.4780 (+0.19 %)	3.4698 (−0.05 %)
Sym1, CC, $b_s = 285.5$, $\beta = 1.5$	3.8130	3.8230 (+0.26 %)	3.8100 (−0.08 %)

The model sizes and computing times are compared in Table 10. With the full basis of 482–569 modes, a GBT analysis, including the cross-section analysis, the pre-buckling solution and the eigenvalue problem, takes 4.6 s on average for the ten cells with $b_s = 150$ mm and $\beta = 1.5$ on two cores of a standard workstation; over all 160 cases of Figures 3 and 4, computed two at a time on the same two cores, it ranged from 2.3 to 23.8 s. The GBT model has about half the unknowns of the FSM and less than half the degrees of freedom of the shell model, but with the full basis it is not faster than the FSM (2.5 s): the global and distortional modes extend over the whole cross-section and couple with all other modes, so that its matrices are less sparse than those of the nodal FSM. The cross-section analysis itself takes about 0.1 s and is reused for all lengths of a parametric study. The efficiency of the GBT formulation therefore lies in its modal description, which provides the mode decomposition and is the basis for reduced mode sets, rather than in the solution time of the full basis.

Table 10. Model size and computing time for $b_s = 150$ mm and $L = 225$ mm.

Model	Unknowns	Computing time per case
Abaqus shell model (S4R/S3, seed 3.7 mm)	approx. 5×10^4 DOFs (estimated from the seed)	[to be added by the authors]
spline finite strip model	37 170 (177 nodal lines × 6 × 35 splines)	2.5 s on average
GBT–Ritz (present)	18 777 (569 modes × 33 splines)	4.6 s on average

The following limitations should be noted. First, GBT and the FSM share the plate assumptions: laminated Reissner–Mindlin walls with a shear correction factor of 5/6, an offset tie represented by stiff link walls or rigid links, and linear buckling about a first-order pre-buckling state. An error common to both, such as the −1 % offset for $b_s = 150$ mm, cannot be detected by comparing them. Second, the periodic condition of Eq. (13), like the constraint equations of the shell model, keeps the width of the cell constant. The Poisson expansion of the cell under the axial load is therefore restrained, and transverse membrane forces S_{ss} arise in the pre-buckling state; for the isotropic cell with $b_s = 150$ mm and $\beta = 3$, omitting S_{ss} from $\mathbf{G}$ would raise the critical load by 23 %. The condition thus represents a panel whose lateral expansion is restrained by the surrounding structure. A periodic condition with free lateral expansion, in which the two side edges differ by a common jump in U_X, requires one additional generalised coordinate and is left for future work.

Third, a cell of width $2b_s$ admits the Bloch wavenumbers $\kappa b_s = 0$ and π only; buckling modes with a longer period across the stringers require a wider cell. Fourth, the full mode basis was used throughout; reduced hierarchical bases of a few tens of modes have not yet been validated for the coupled lay-ups with clamped loaded edges.

4. Conclusions

This study presented a shear-deformable GBT–Ritz formulation for the buckling analysis of omega-stringer-stiffened composite panels in which the periodic boundary conditions of the panel cell are built into the cross-section analysis. The conventional deformation modes are obtained by projecting out the constraint rows of the periodic condition family by family, the warping of the global and distortional modes follows from a constrained Vlasov least-squares problem, and the shear and Mindlin modes are obtained by merging the nodal functions of the two side edges. Every mode, and therefore every combination and truncation of the modes, satisfies the periodic condition to machine precision, so that no penalty terms or Lagrange multipliers are needed in the member analysis. The modal matrices were given explicitly in the notation of classical GBT and of the GBT–Ritz approach for symmetric and unsymmetric laminates, and the formulation reduces to the classical GBT buckling equations and the classical geometric matrices for uncoupled walls under uniform compression, which was confirmed numerically to 0.001 %. For 30 panels with simply supported loaded edges, the critical loads agree with Abaqus shell models with periodic constraint equations within 0.57 % on average and 1.25 % at most, and for 30 panels with clamped loaded edges they agree with an independent spline finite strip model within 0.30 % on average and 1.17 % at most. The accuracy is the same for cross-ply, angle-ply and unsymmetric laminates, which confirms the consistent treatment of the bending–twisting and membrane–bending coupling.

Beyond accuracy, the formulation provides insight into the buckling mechanisms of the cell. The energy-based mode decomposition identifies the critical modes of all panels studied as local skin buckling, with global and distortional contributions below 4 %, and the variation of the critical load with the aspect ratio of the cell follows directly from the same model at a computing time of a few seconds per case. The periodic condition removes the arbitrariness of the side-edge modelling of panel cells, and its formulation within the modes makes it available to reduced GBT mode bases. Future work will address the confirmation of the clamped cases with shell finite element models, periodic conditions with free lateral expansion of the cell, reduced hierarchical mode bases for parametric studies and surrogate models, and the comparison with the closed-form periodic solutions of El Yaakoubi-Mesbah and Mittelstedt [17].

Author Contributions: Conceptualization, A.M.E. and N.K.; methodology, A.M.E. and N.K.; software, A.M.E.; validation, A.M.E.; formal analysis, A.M.E.; investigation, A.M.E.; writing—original draft preparation, A.M.E.; writing—review and editing, N.K. and C.M.; supervision, N.K. and C.M.; project administration, C.M.; funding acquisition, C.M. All authors have read and agreed to the published version of the manuscript. [To be confirmed by the authors.]

Funding: [To be completed by the authors.]

Data Availability Statement: The data presented in this study are available on request from the corresponding author.

Conflicts of Interest: The authors declare no conflicts of interest.

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